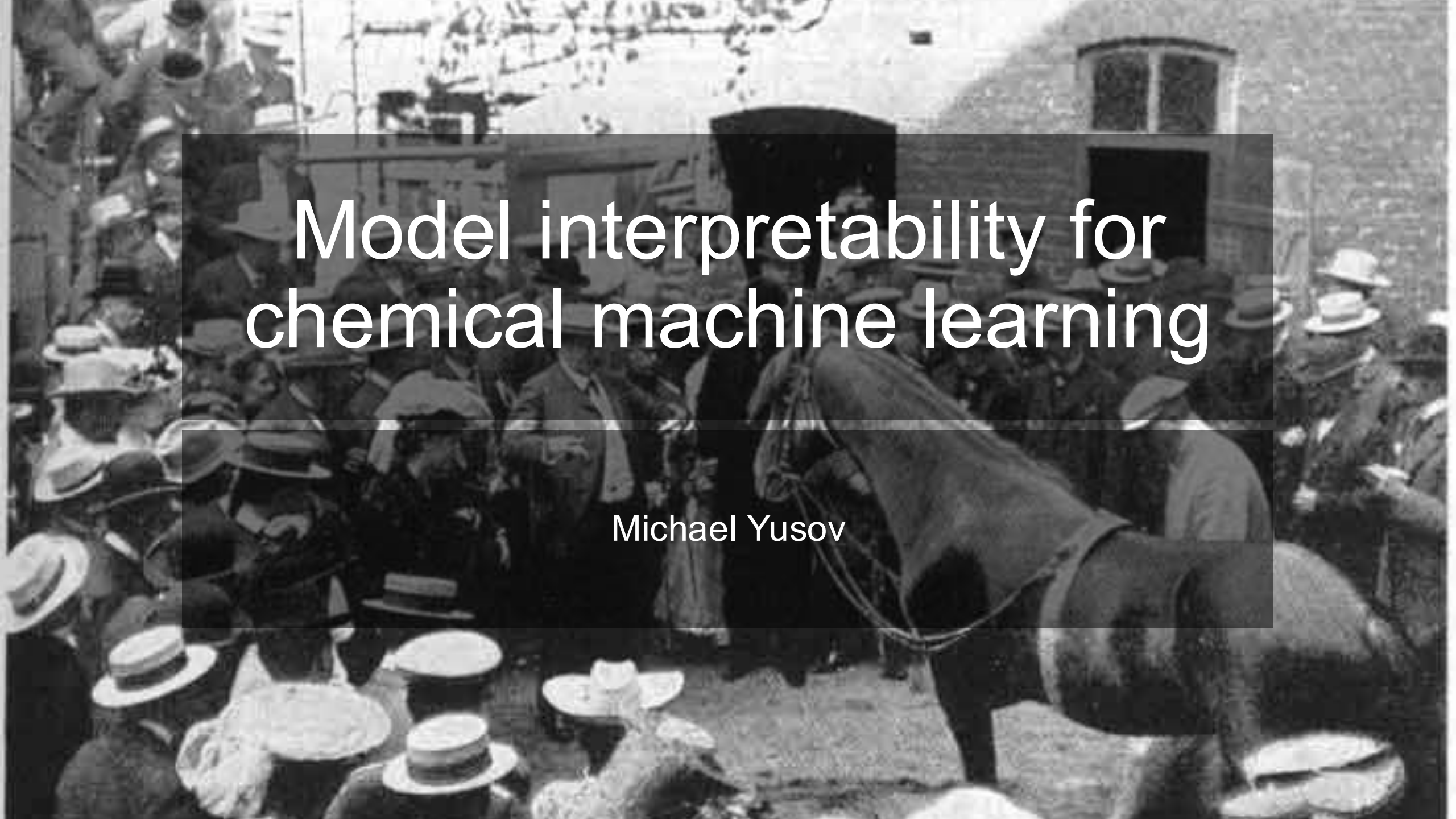


A black and white photograph of a crowded outdoor event, possibly a fair or festival. In the foreground, many people are visible, mostly wearing hats, suggesting a sunny day. The crowd is dense, and the background shows a large, dark structure, possibly a tent or a building, with a small arched window. The overall scene is lively and busy.

Model interpretability for chemical machine learning

Michael Yusov

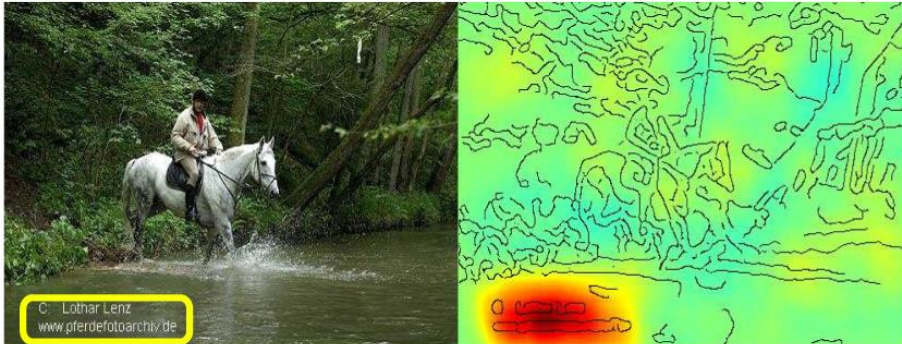
Clever Hans

“In the field of artificial intelligence, **the Clever Hans effect** describes a phenomenon where an algorithm seems to make correct predictions without having the relevant data and/or by using incorrect reasoning.” —Wikipedia



The Clever Hans effect in AI

Horse-picture from Pascal VOC data set



Source tag
present



Classified
as horse

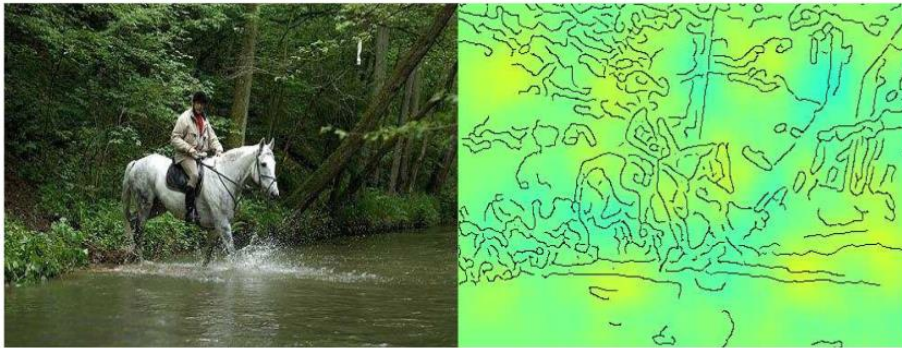
Artificial picture of a car



No source
tag present



Not classified
as horse



What is model interpretability?

Definition: Model interpretability is the extent to which a human can understand how a machine learning model uses input features to produce predictions, including why a particular prediction was made and what relationships the model has learned.

In this lecture, we use “interpretability” and “explainability” interchangeably.

Why study model interpretability for chemical machine learning?

We study interpretability to:

1. avoid blindly trusting predictive performance and check whether the model learned meaningful features,
2. generate scientific insight and make predictions actionable,
3. build trust with collaborators or debug models

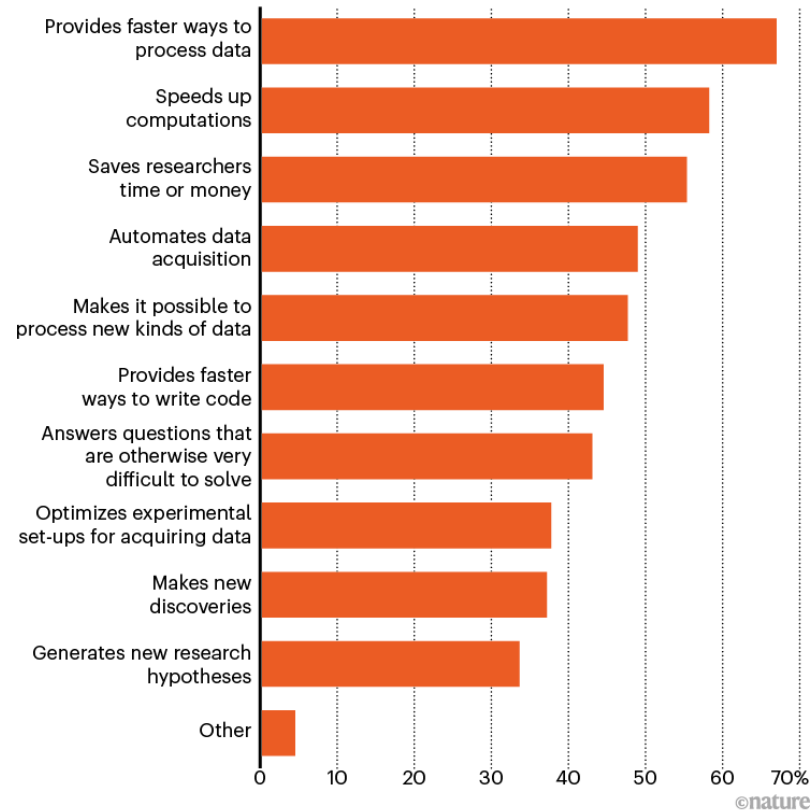
The goal of science is to gain knowledge; knowing why a model made certain predictions can help us learn more about the chemistry we are studying.

Interpretable machine learning allows the model itself to become the source of knowledge instead of the data.

“AI and science: what 1,600 researchers think” (September 2023)

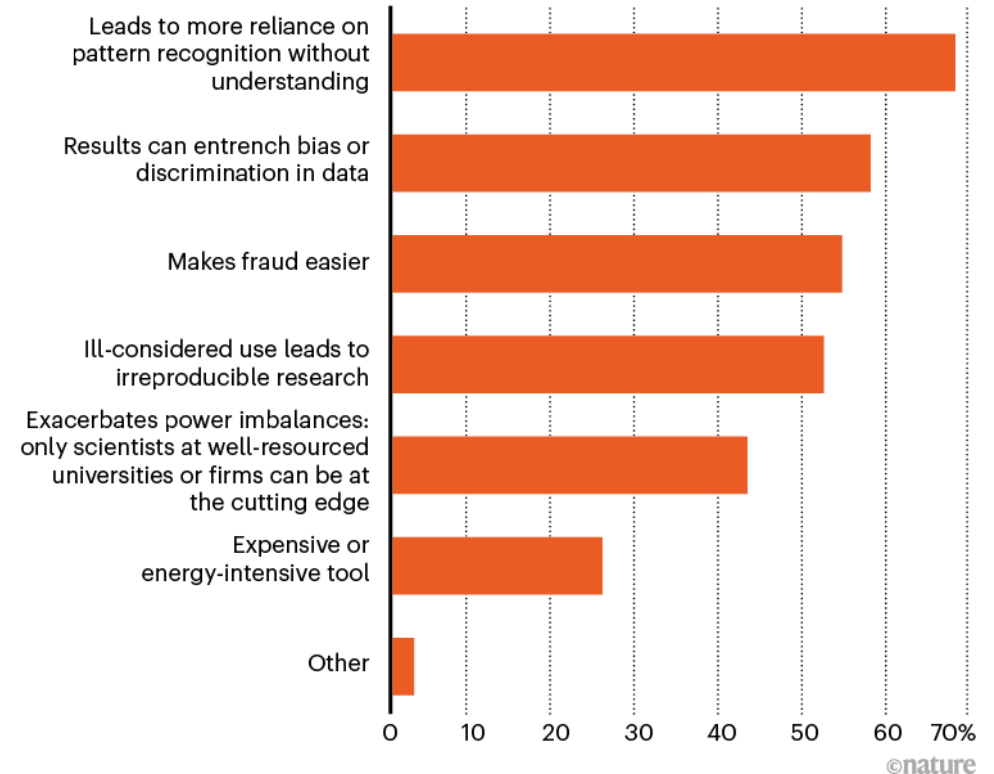
POSITIVE IMPACTS OF AI

Q: Considering machine-learning methods, what do you think are positive impacts of AI in research? (Choose all that apply.)

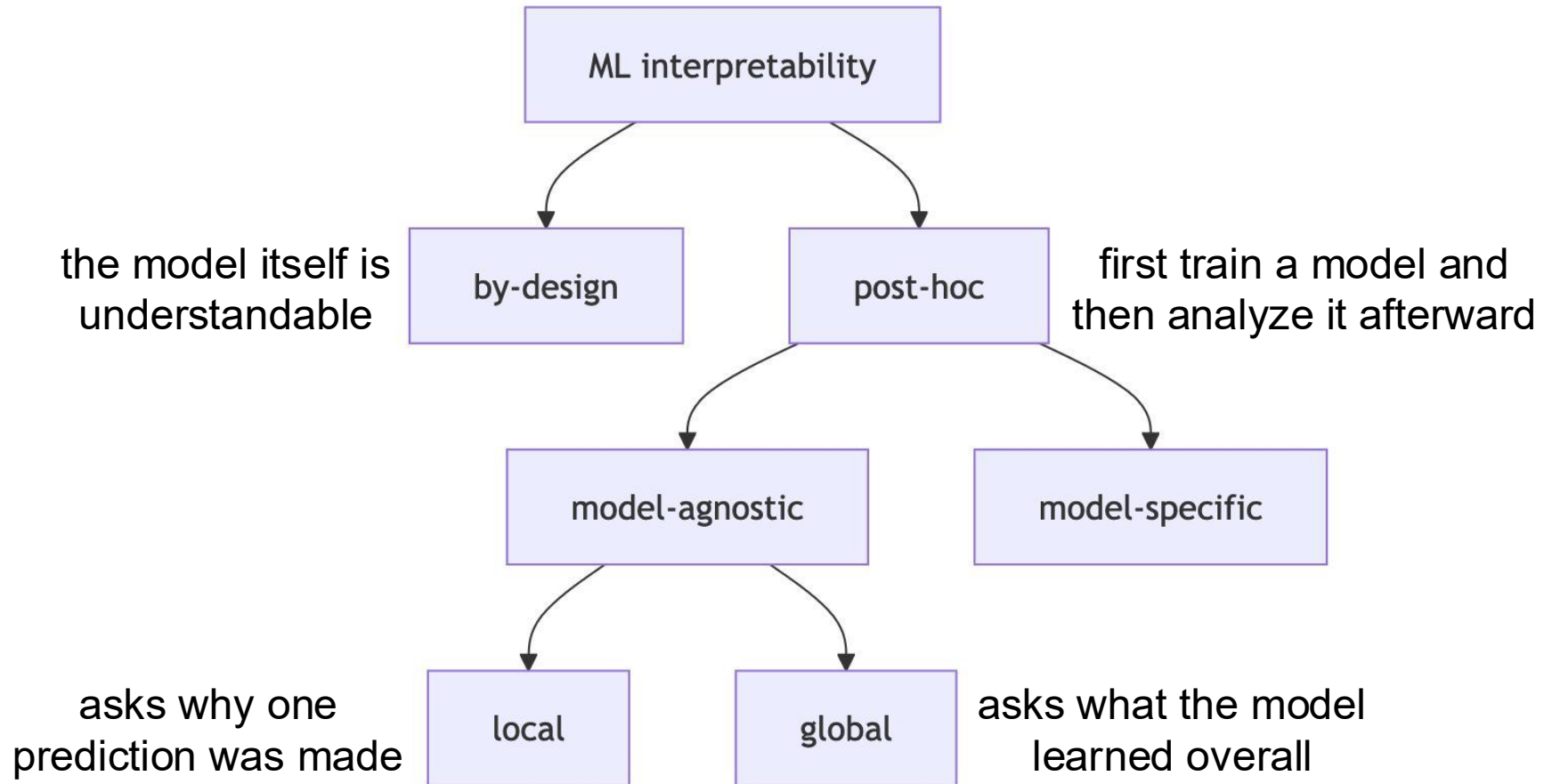


NEGATIVE IMPACTS OF AI

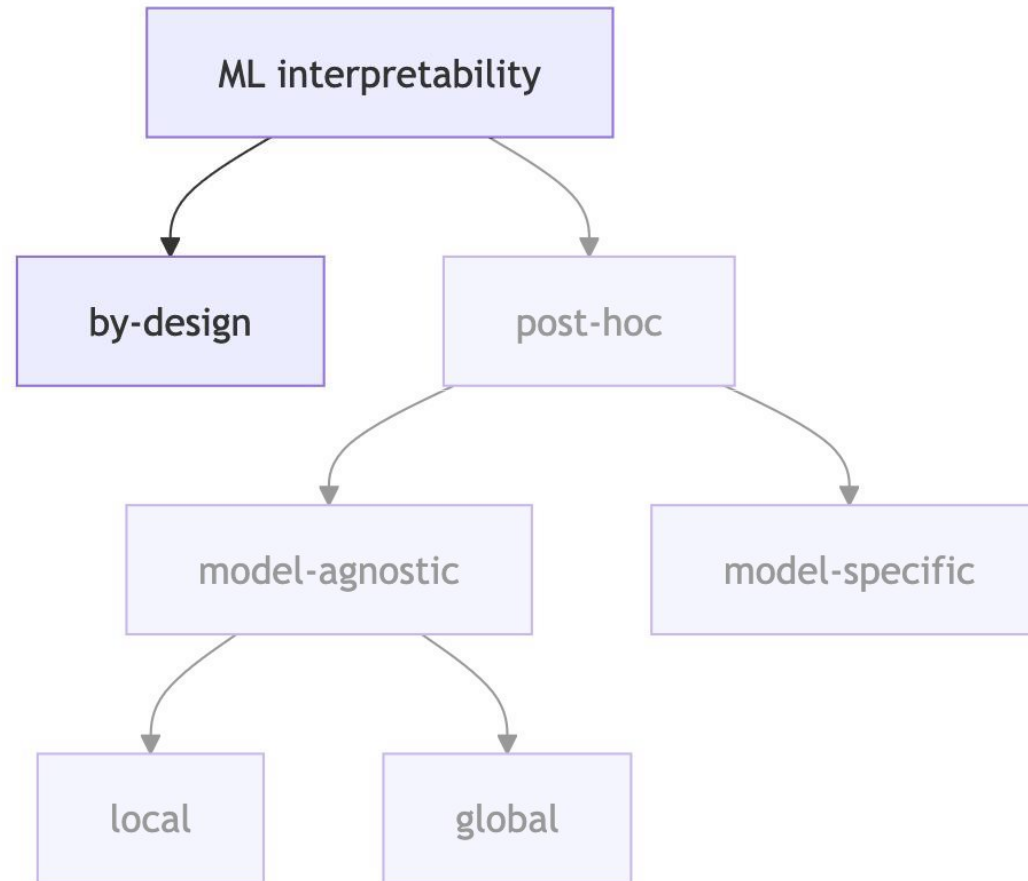
Q: Considering machine-learning methods, what do you think are negative impacts of AI in research? (Choose all that apply.)



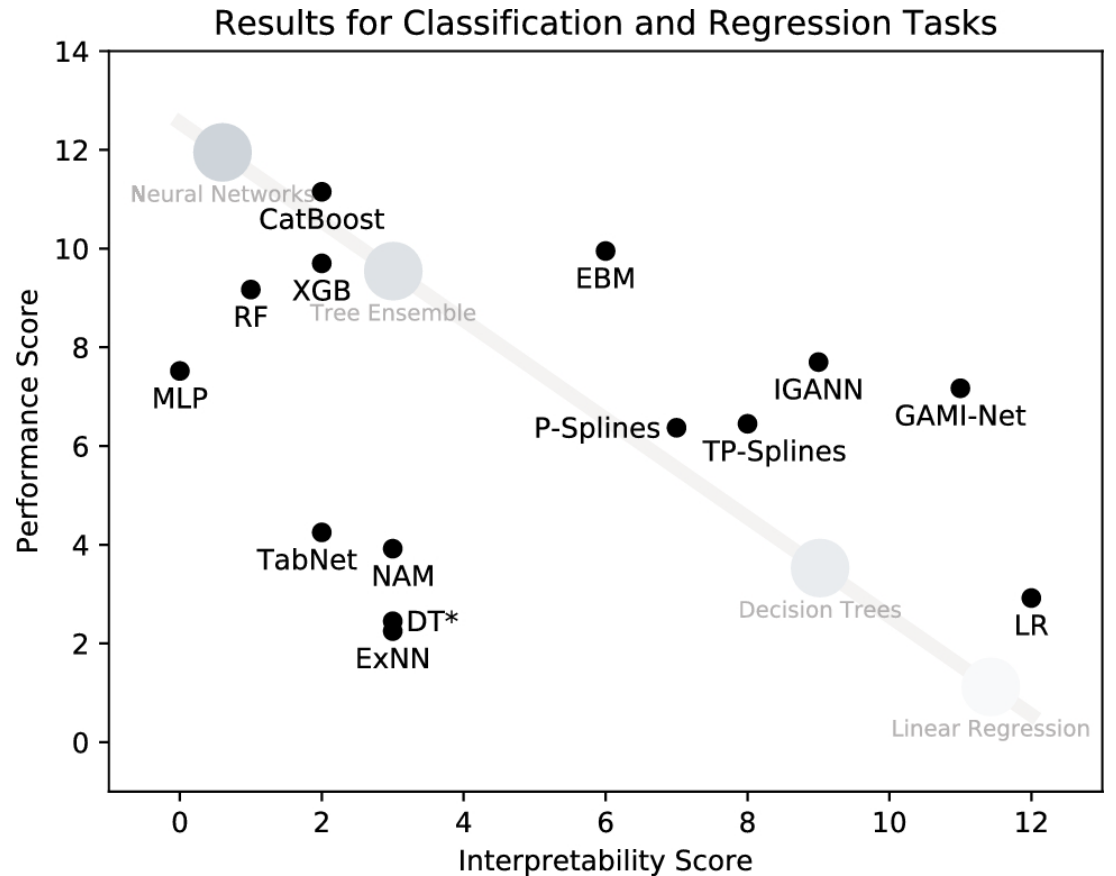
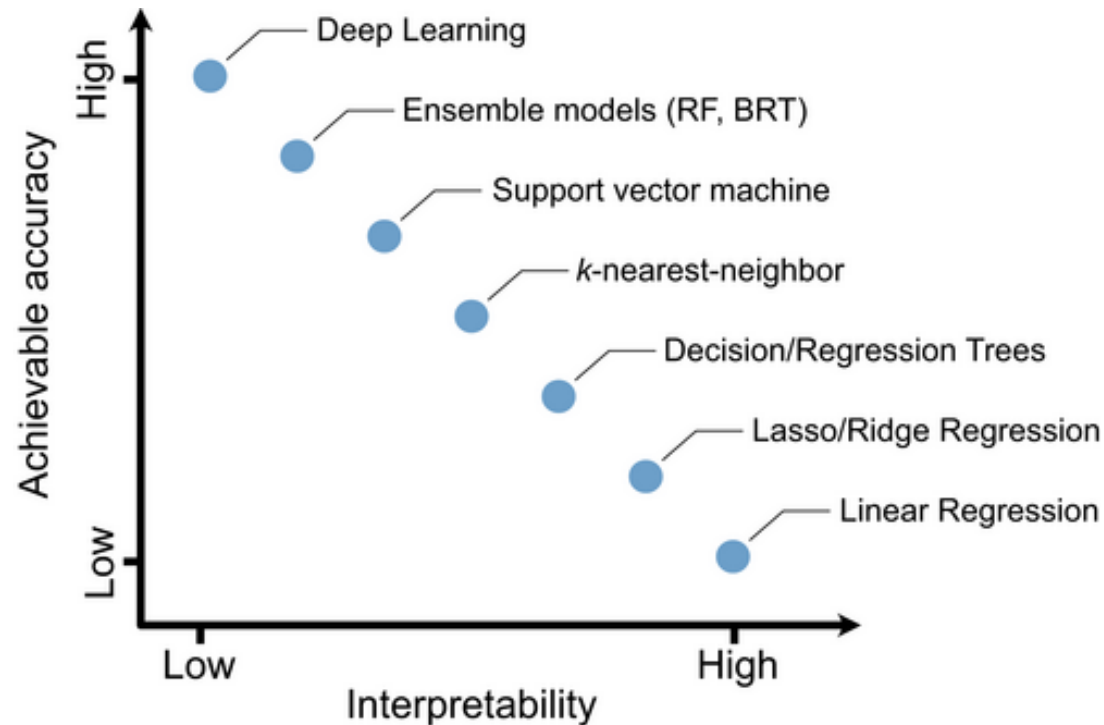
Taxonomy of interpretability methods



Interpretability by design



A common misconception: the performance-interpretability trade-off

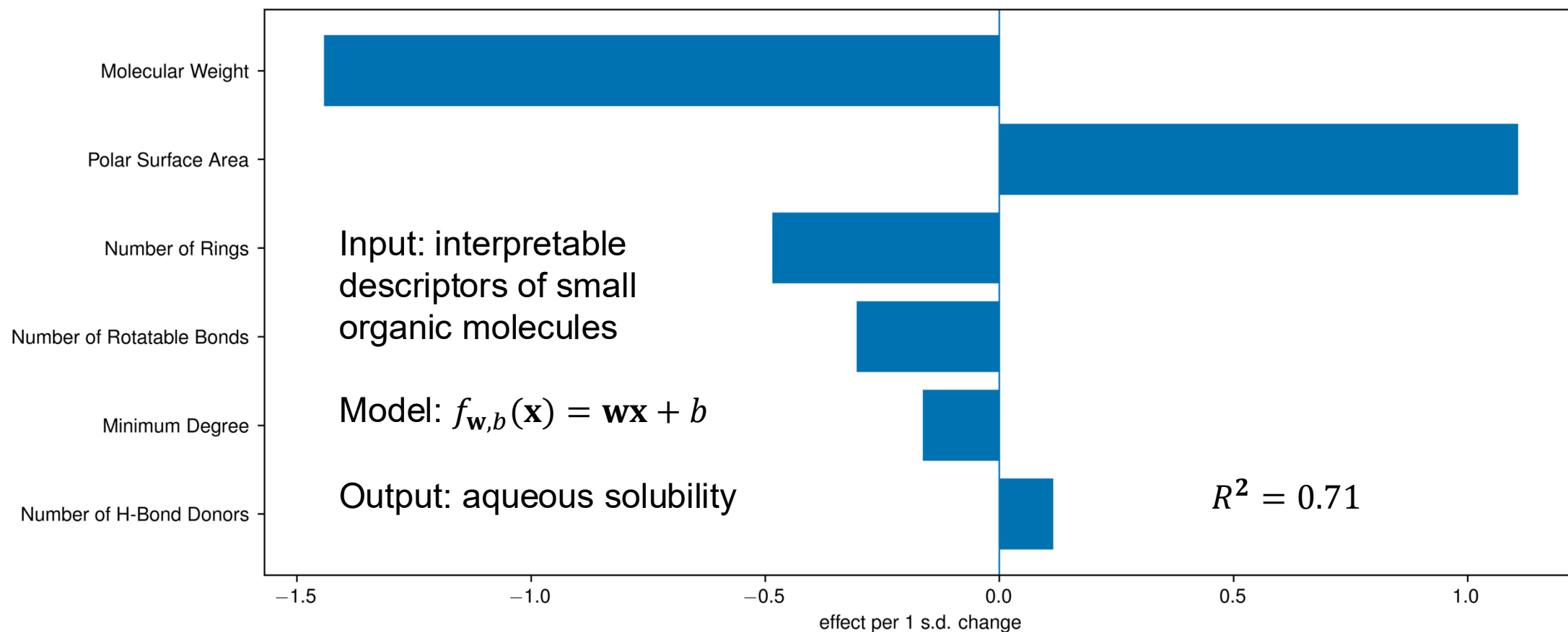


Pichler, M. & Hartig, F. Machine learning and deep learning—A review for ecologists. *Methods Ecol. Evol.* 14, 994–1016 (2023).

<https://doi.org/10.1111/2041-210X.14061>

Kruschel, S. *et al.* Challenging the performance-interpretability trade-off: An evaluation of interpretable machine learning models. *Bus. Inf. Syst. Eng.* **68**, 9 159–183 (2026). <https://doi.org/10.1007/s12599-024-00922-2>

Interpretability by design: linear model coefficients

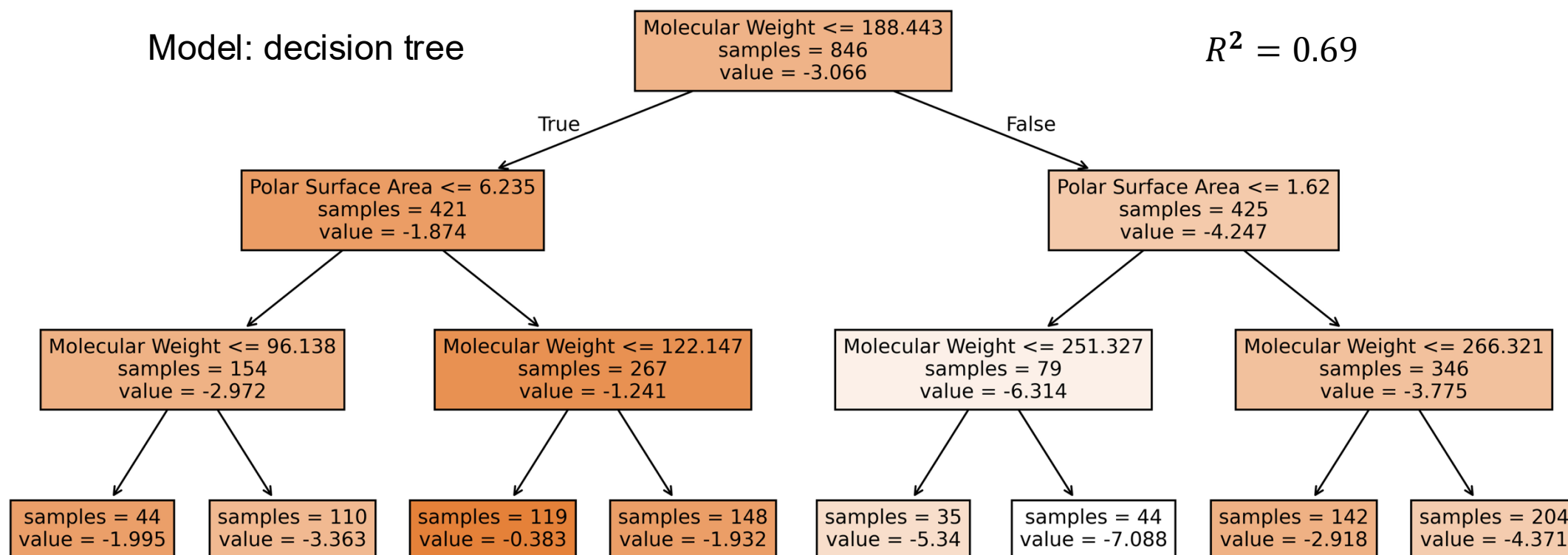


We will use the Delaney/ESOL aqueous solubility dataset for this lecture

Interpretability by design: decision trees

Model: decision tree

$R^2 = 0.69$



Interpretability by design

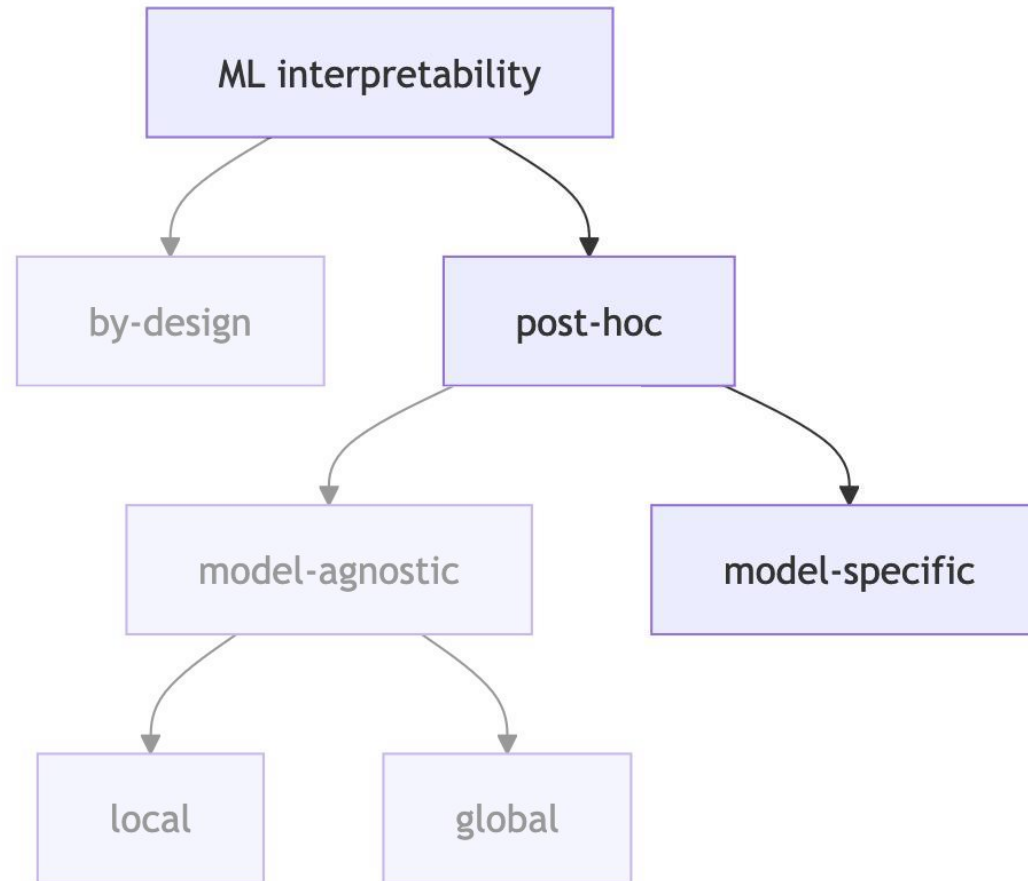
Strengths

- **Provides directly inspectable models** whose structure is understandable from the start
- **Helps reveal** simple trends, signs, rules, or constraints

Limitations

- **Does not guarantee high predictive accuracy** if the true relationship is highly nonlinear or interaction-rich
- **Can mislead when** simple coefficients or rules are treated as mechanistic truth

Model-specific post-hoc interpretability



Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

Definition: The impurity of a node following the squared-error criterion is the variance of the target values in that node.

$$I(t) = \frac{1}{N_t} \sum_{i \in t} (y_i - \bar{y}_t)^2$$

where:

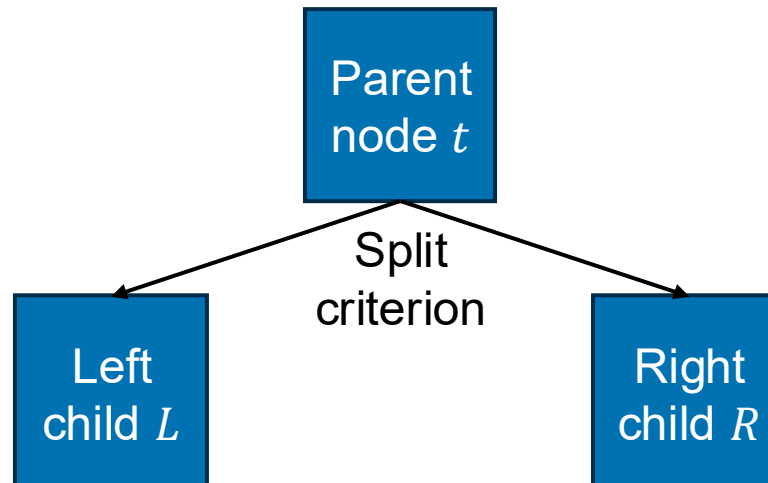
- $I(t)$: impurity/within-node variance of node t
- N_t : number of training samples in node t
- y_i : measured target value of sample i
- $\bar{y}_t$: average target value of samples in node t

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

$$\Delta I(t) = \frac{N_t}{N} \left[I(t) - \frac{N_L}{N_t} I(L) - \frac{N_R}{N_t} I(R) \right]$$

where:

- N : total number of training samples in the tree
- N_L, N_R : number of samples in the left and right children



Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

For one tree, the importance of feature j is:

$$\text{importance}_j = \frac{\sum_{t: \text{split feature}(t)=j} \Delta I(t)}{\sum_k \sum_{t: \text{split feature}(t)=k} \Delta I(t)}$$

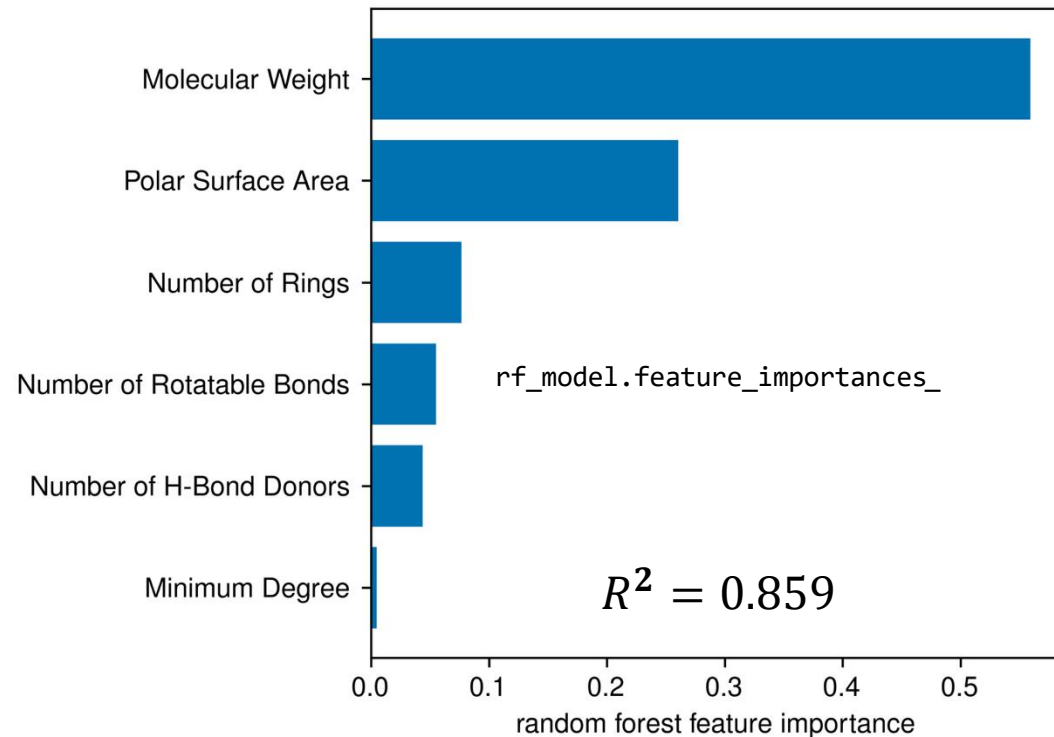
For random forests and gradient boosted tree models:

$$\text{RF/GB importance}_j = \frac{1}{T} \sum_{i=1}^T \text{importance}_j^{(m)}$$

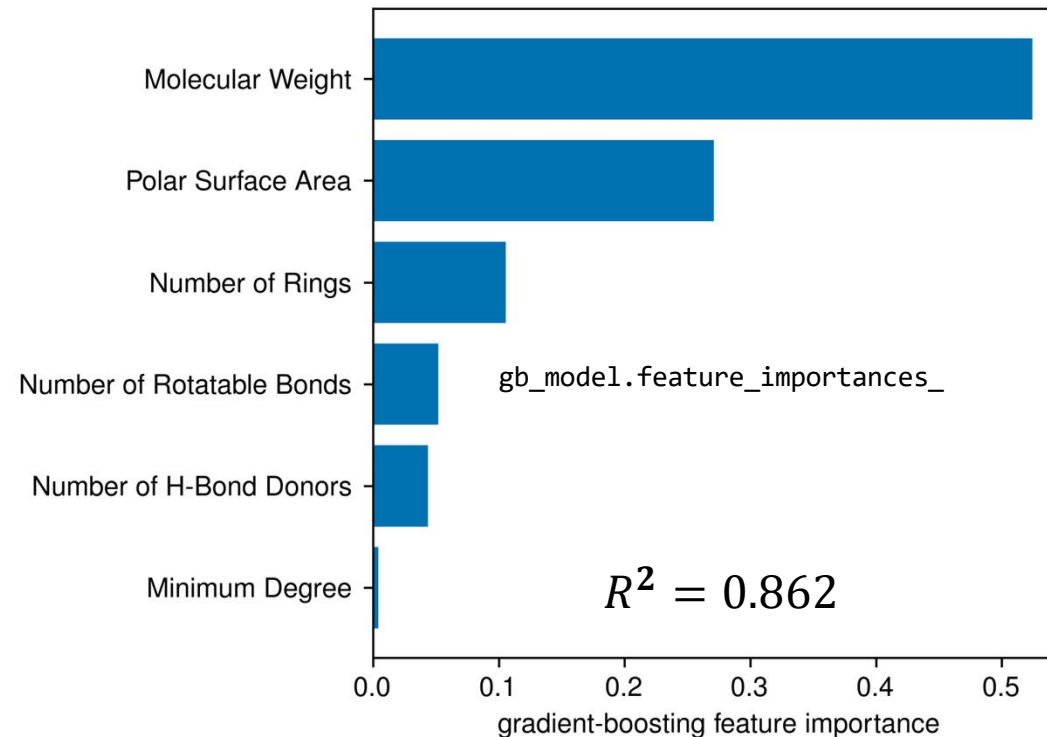
where $\text{importance}_j^{(m)}$ is the importance of feature j in tree m .

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

Random forest



Gradient boosted tree



Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

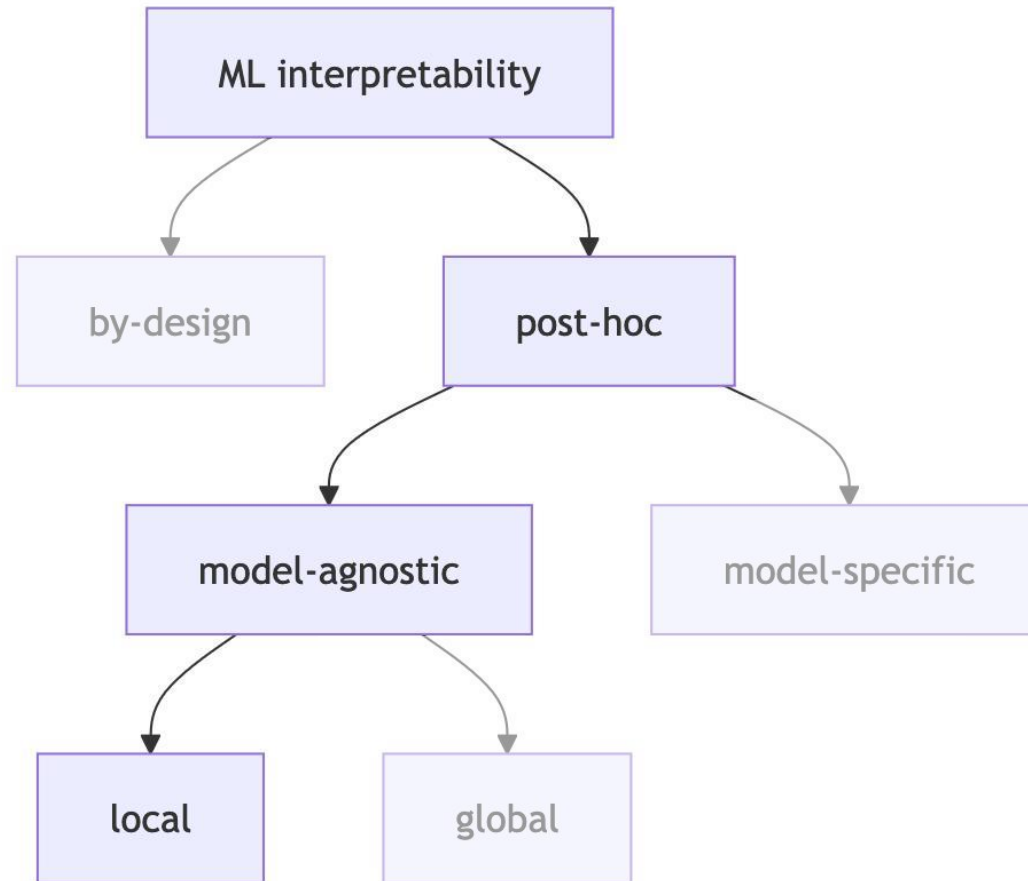
Strengths

- **Provides a fast global feature ranking** for random forests and gradient boosted trees
- **Works well when** you want a quick first look at which features helped trees make useful splits

Limitations

- **Does not show direction or local effects** because importances are unsigned and global
- **Sensitive to feature correlation** because correlated features can receive biased/split importance

Local post-hoc model-agnostic local interpretability



Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

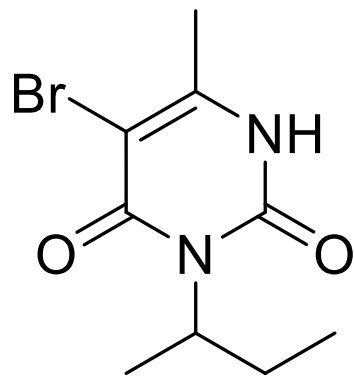
Core idea: Explain an individual prediction by fitting a simple model that mimics a complex model only near the point being explained.

Let the original model be $f(x)$ and define a simple, typically linear surrogate model $g(z)$. The goal of LIME is for g to behave like f near the data point being evaluated.

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 1: Pick the molecule to explain



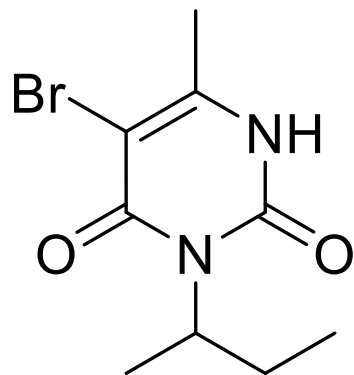
Bromacil

$$f_{gb}(x_i) = -3.252$$

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 2: Generate perturbed versions of the molecule



Bromacil

$$f_{gb}(x_i) = -3.252$$

Descriptor values:

Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area
1	261.119	1	1	2	54.86

Generate 5000 perturbed samples nearby:

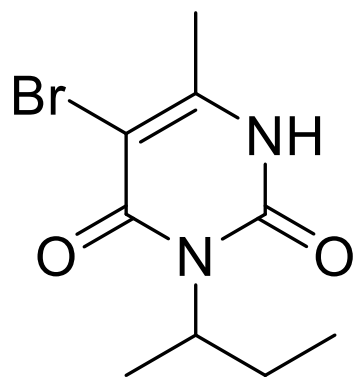
Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area
0.997	69.567	3.301	2.000	2.564	39.467
0.911	226.866	3.222	3.064	9.390	0.000
0.918	455.888	0.000	4.111	2.352	95.249

⋮

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 3: Predict the perturbed samples with f



Bromacil

$$f_{gb}(x_i) = -3.252$$

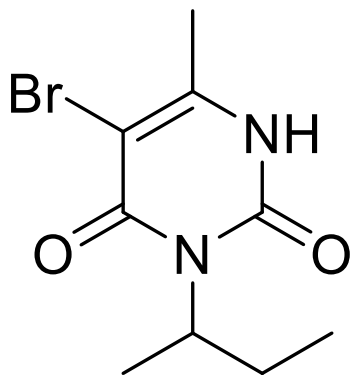
Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area	$f(z)$
0.997	69.567	3.301	2.000	2.564	39.467	-0.060
0.911	226.866	3.222	3.064	9.390	0.000	-8.016
0.918	455.888	0.000	4.111	2.352	95.249	-5.524

⋮

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 4: Compute how close each perturbed sample is to the original molecule



Bromacil

$$f_{gb}(x_i) = -3.252$$

Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area	$f(z)$	$\pi_{x_i}(z)$
0.997	69.567	3.301	2.000	2.564	39.467	-0.060	0.641
0.911	226.866	3.222	3.064	9.390	0.000	-8.016	0.553
0.918	455.888	0.000	4.111	2.352	95.249	-5.524	0.553

⋮

A common choice of kernel is:

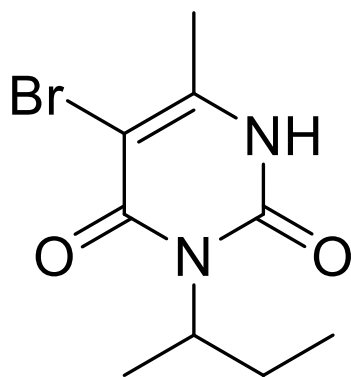
$$\pi_{x_i}(z) = \exp\left(-\frac{D(x_i, z)^2}{\sigma^2}\right)$$

- $\pi_{x_i}(z)$: weight assigned to perturbation sample z
- $D(x_i, z)$: Euclidean distance
- σ : kernel width

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 5: Fit a simple weighted surrogate model



Bromacil

$$f_{gb}(x_i) = -3.252$$

$$g(x_i) = -2.809$$

$$R^2 = 0.054$$

$$g(z) = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_p z_p$$

$$\xi(x_i) = \underset{g \in G}{\operatorname{argmin}} \sum_{k=1}^{5000} \pi_{x_i}(z_k) [f(z_k) - g(z_k)]^2$$
$$\|\beta\|_0 \leq 4$$

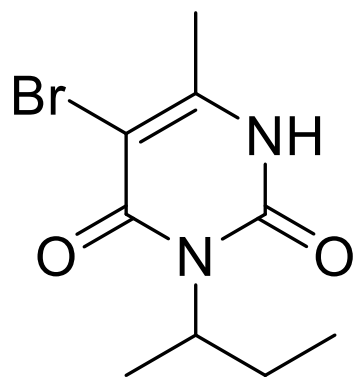
Using rule-based bins instead of feature coefficients,

$$g(z) = -3.279$$
$$+ 1.099 * I[26.30 < \text{Polar Surface Area} \leq 56.79]$$
$$- 0.685 * I[190.35 < \text{Molecular Weight} \leq 278.06]$$
$$+ 0.151 * I[1.00 < \text{Number of Rotatable Bonds} \leq 3.00]$$
$$- 0.067 * I[\text{Minimum Degree} \leq 1.00]$$

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 5: Fit a simple weighted surrogate model

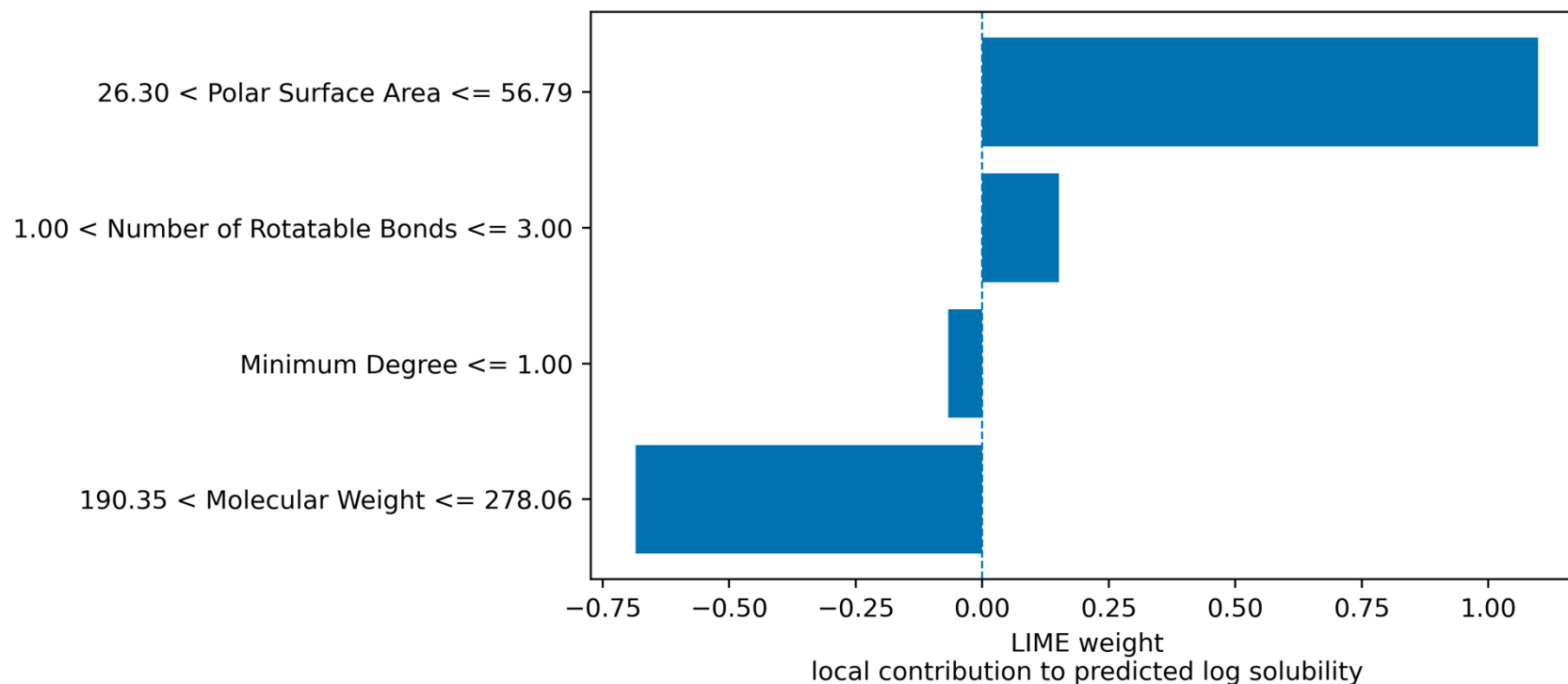


Bromacil

$$f_{gb}(x_i) = -3.252$$

$$g(x_i) = -2.809$$

$$R^2 = 0.054$$



Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

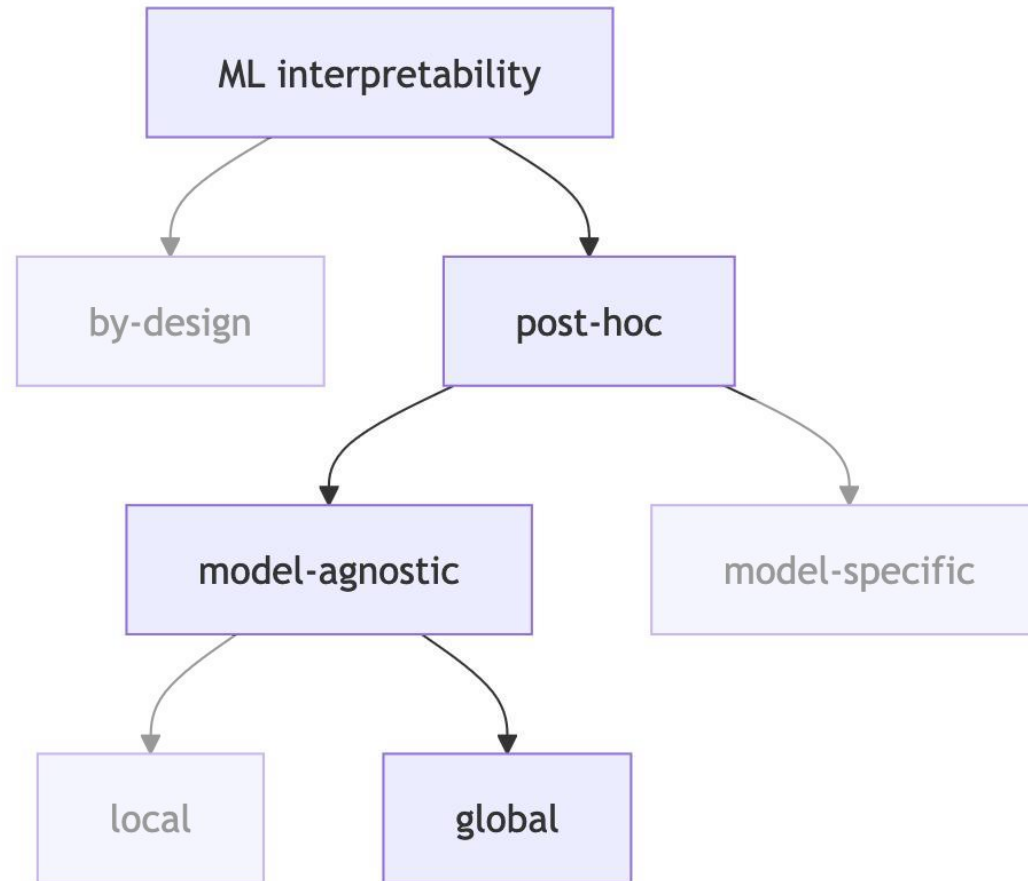
Strengths

- **Provides local, human-readable explanations** for individual predictions
- **Works well when** a simple local surrogate can approximate the black-box model near the sample being explained

Limitations

- **Does not explain the black-box model directly**; it explains a local surrogate approximation
- **Sensitive to perturbation strategy, random seed, kernel width, number of features, etc.**
- **Can mislead when** the local surrogate has poor fidelity

Global post-hoc model-agnostic local interpretability



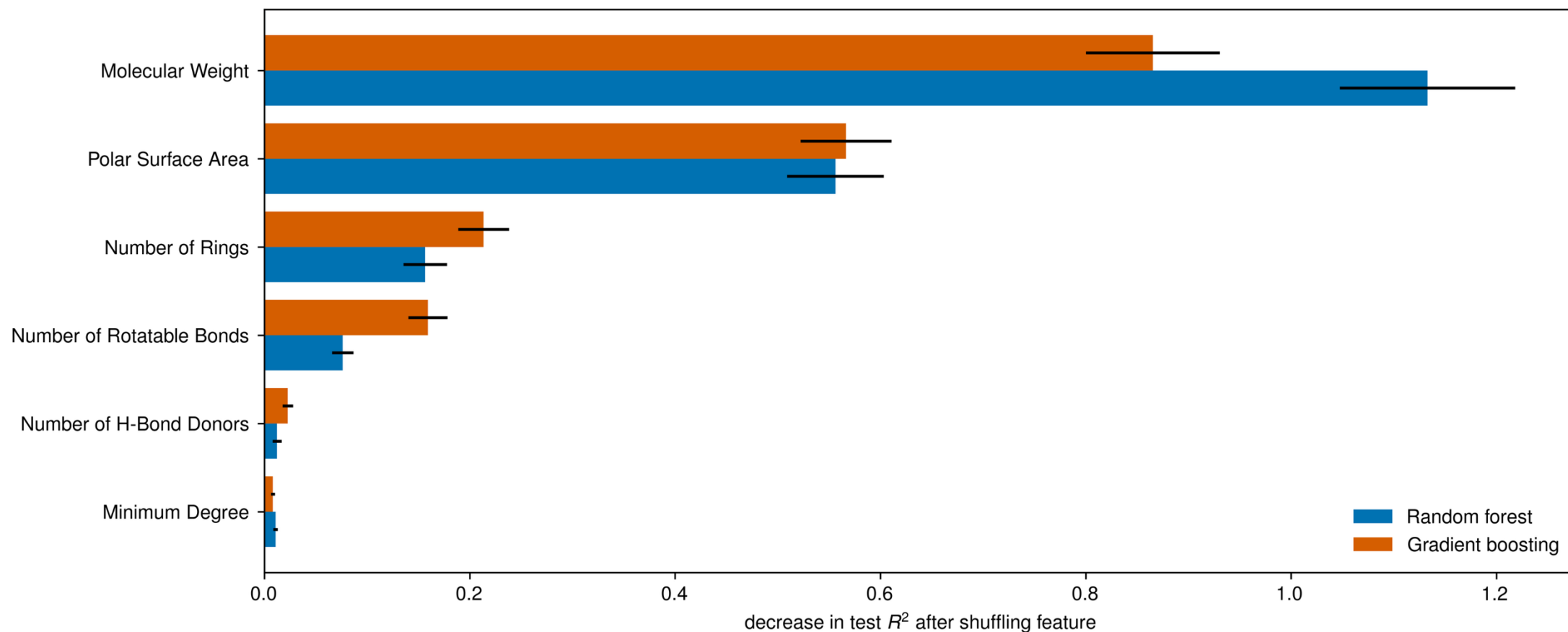
Global model-agnostic post-hoc interpretability: permutation importance

Let f be a fitted model, let X be the dataset used for inspection (usually a held-out validation or test set), and let y be the corresponding target values.

1. Compute the baseline score: $s = \text{score}(f, X, y)$ (here, we will use $s = R^2(y, f(X))$).
2. Choose feature j . Randomly permute column j , leaving all other columns unchanged. Call the corrupted dataset X_j .
3. Compute the score on the corrupted dataset: $s_j = \text{score}(f, X_j, y)$
4. The permutation importance of feature j is: $\text{PI}_j = s - s_j$.
5. Repeat K times for each feature and average the score after shuffling:

$$\text{PI}_j = s - \frac{1}{K} \sum_{k=1}^K s_{k,j}$$

Global model-agnostic post-hoc interpretability: permutation importance



Global model-agnostic post-hoc interpretability: permutation importance

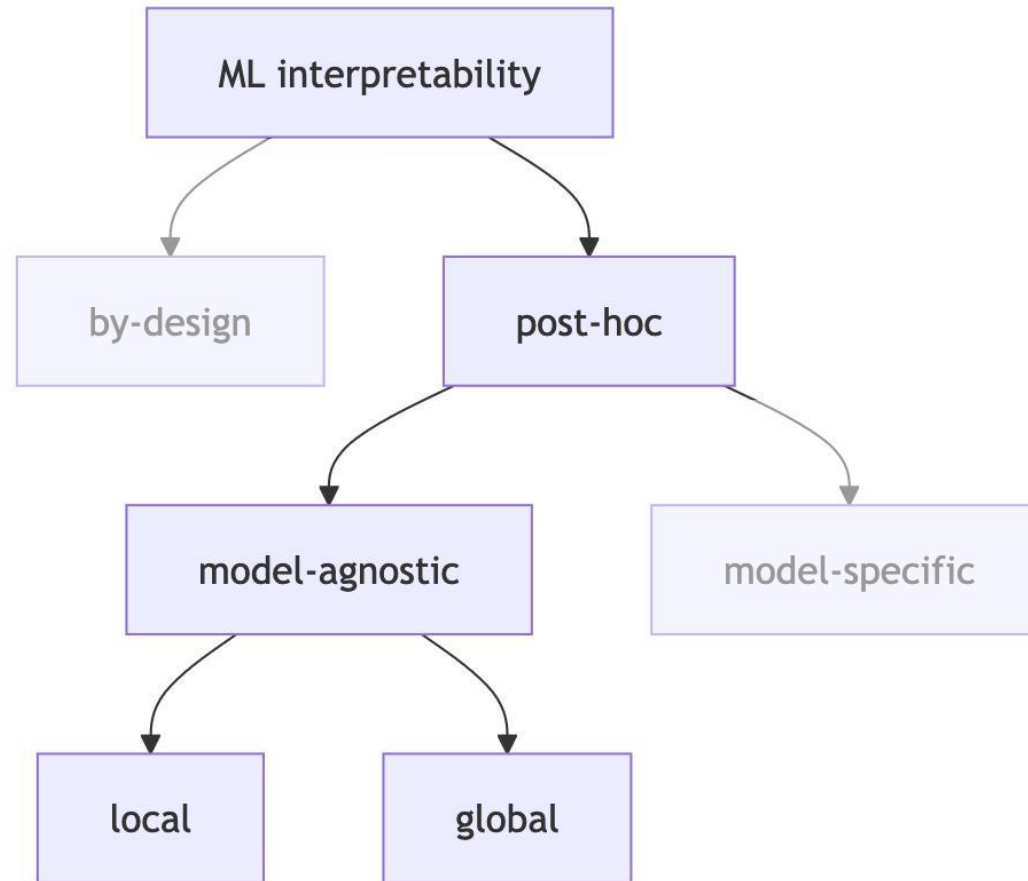
Strengths

- **Provides a model-agnostic global feature ranking** based on predictive performance
- **Helps reveal** how much performance decreases when each feature's information is destroyed

Limitations

- **Does not show direction or local effects** because it measures performance drop, not positive/negative contribution
- **Can mislead when** redundant descriptors hide each other's importance

Local & global post-hoc model-agnostic local interpretability



Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)

Let $f(x)$ be a fitted model.

For measurement i , feature j , and a chosen value z for $x_{i,j}$, the ICE value is:

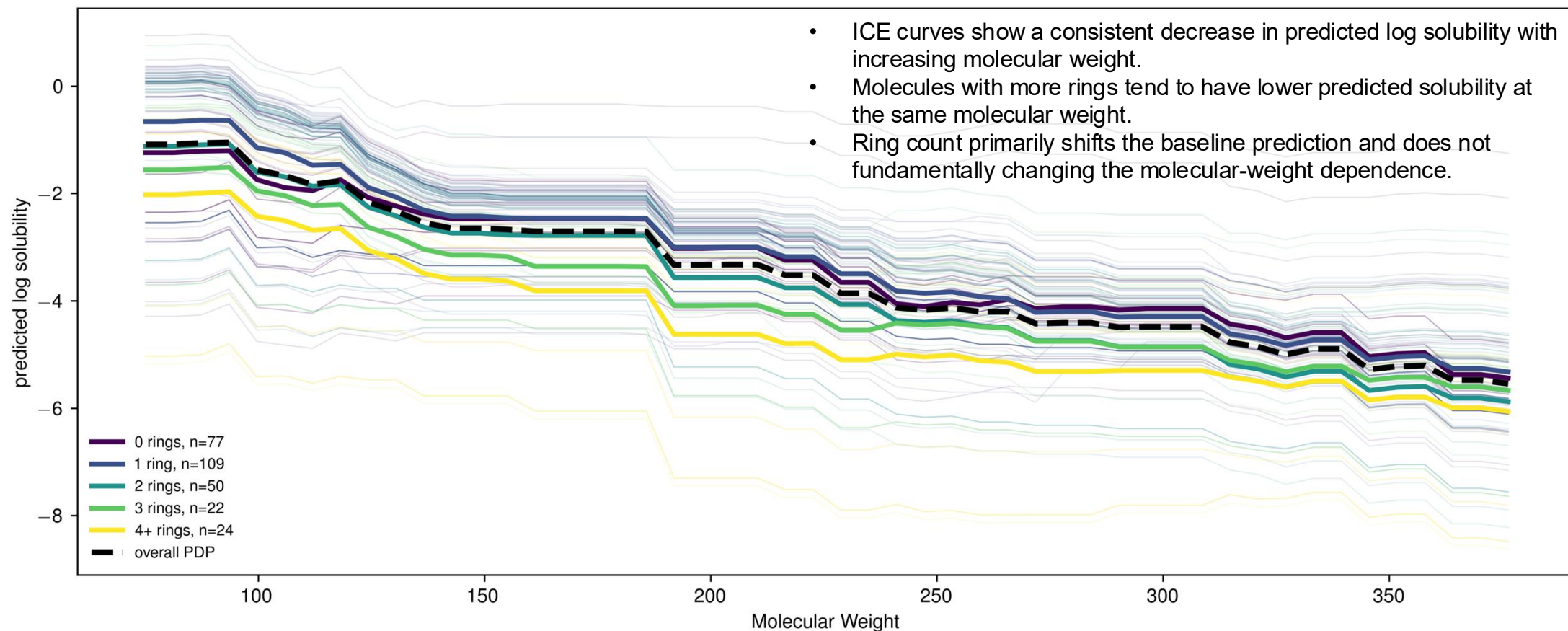
$$\text{ICE}_{i,j}(z) = f(z, x_{i,-j})$$

$x_{i,-j}$: all features other than j for measurement i

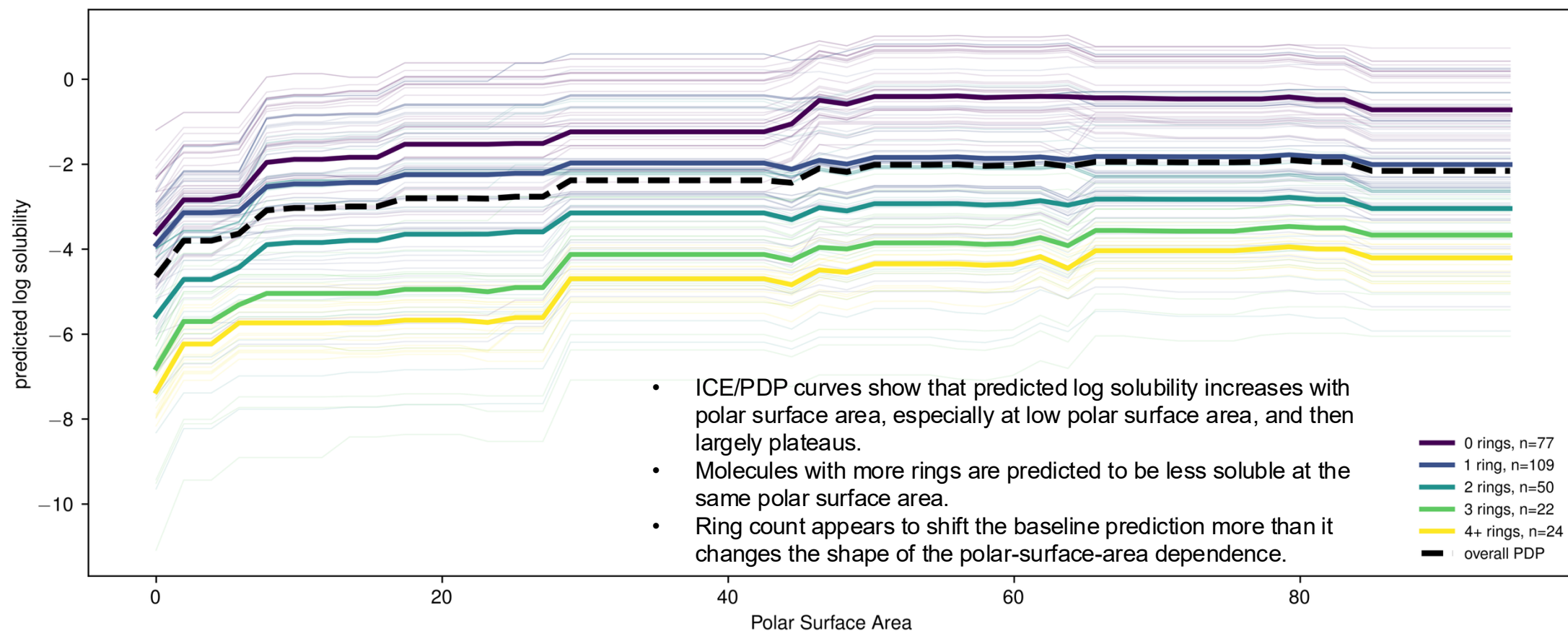
A PDP averages the ICE curves over all measurements:

$$\text{PDP}_j(z) = \frac{1}{n} \sum_{i=1}^n f(z, x_{i,-j})$$

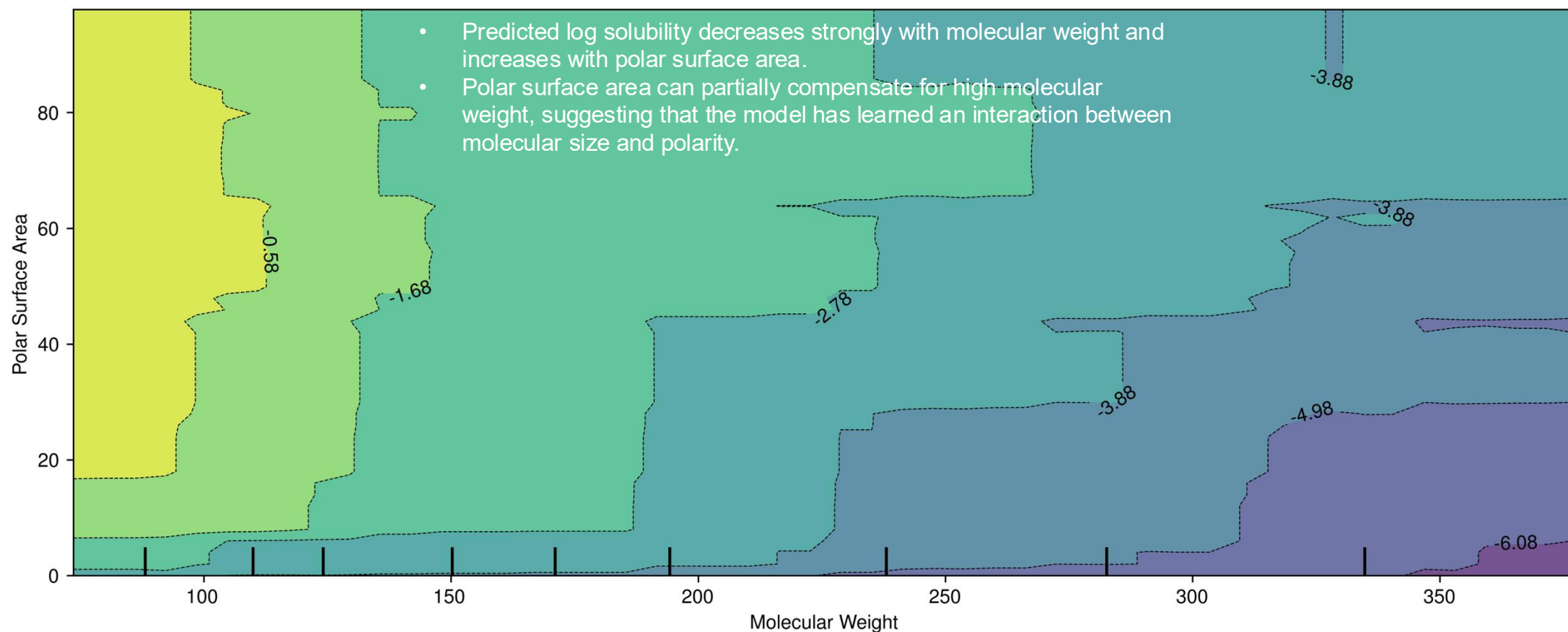
Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)



Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)



Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)



Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)

Strengths

- **Works well when** you want to move from “which features matter?” to “how does the model use this feature?”
- **Helps reveal** direction, nonlinearity, thresholds, saturation, and heterogeneity across samples

Limitations

- **Do not prove causal effects** because they evaluate model behavior under artificial feature changes
- **Can mislead when** PDP averages hide heterogeneous ICE behavior

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Core idea: every prediction is decomposed as

$$f(x_i) = \varphi_0 + \varphi_{i,1} + \varphi_{i,2} + \cdots + \varphi_{i,p}$$

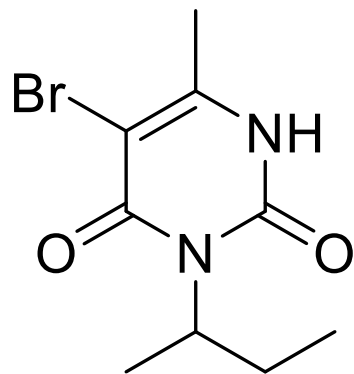
where

- $f(x_i)$: model prediction for sample i
- φ_0 : baseline prediction
- $\varphi_{i,j}$: SHAP value for feature j in sample i
- p : number of features

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

$$f_{gb}(x_i) = -3.252$$



Bromacil

$$\varphi_0 = \mathbb{E}[f(X)] = -2.885$$

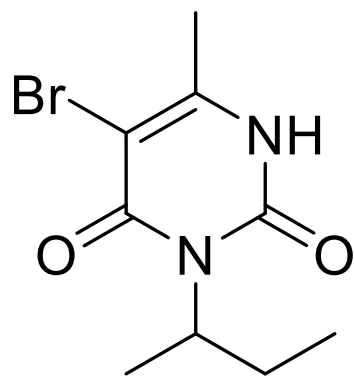
$$\begin{aligned} \therefore \sum \varphi_{i,j} &= \varphi_{i,\text{mw}} + \varphi_{i,\text{psa}} + \varphi_{i,\text{\#r}} + \varphi_{i,\text{\#hbd}} + \varphi_{i,\text{md}} + \varphi_{i,\text{\#rb}} \\ &= -3.252 - (-2.885) = -0.367 \end{aligned}$$

How much did molecular weight, polar surface area, ring count, and the other descriptors push the predicted log solubility above or below a reference prediction?

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

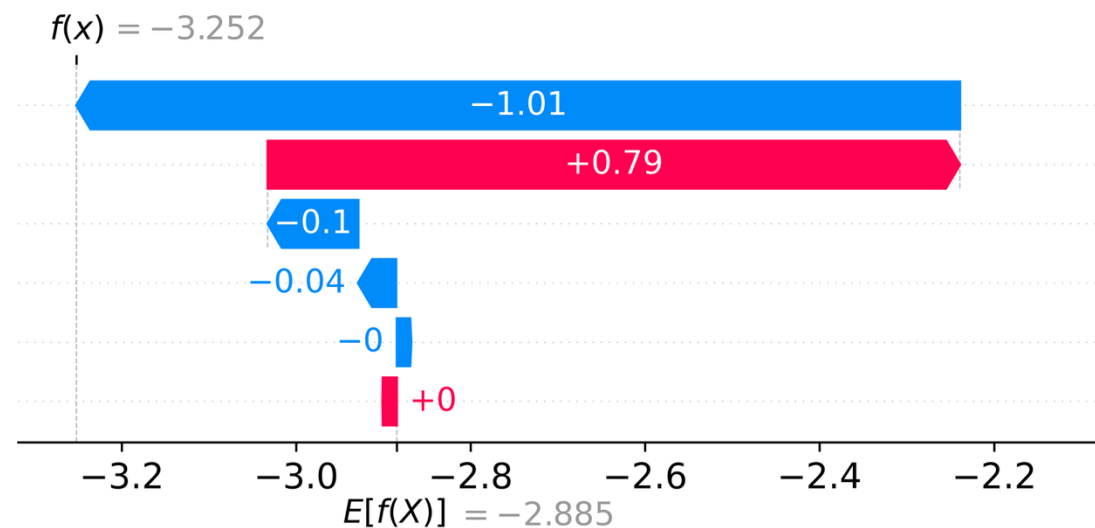
Example:

$$f_{gb}(x_i) = -3.252$$



Bromacil

261.119 = Molecular Weight
54.86 = Polar Surface Area
1 = Number of Rings
1 = Number of H-Bond Donors
1 = Minimum Degree
2 = Number of Rotatable Bonds

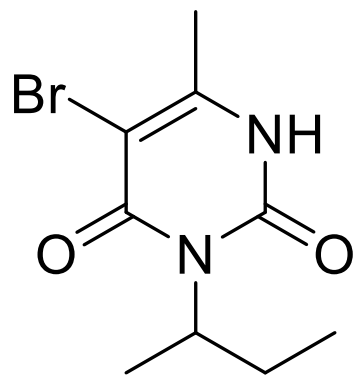


How much did molecular weight, polar surface area, ring count, and the other descriptors push the predicted log solubility above or below a reference prediction?

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 1: Pick the molecule to explain



Bromacil

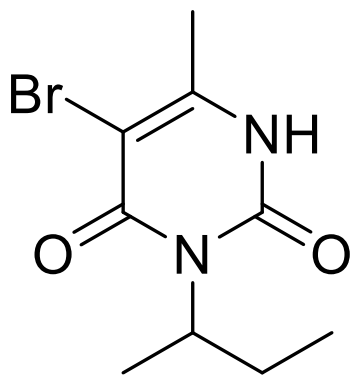
$$f_{gb}(x_i) = -3.252$$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 2: Choose a background dataset X

$X = x_{\text{train}}$



Bromacil

$$f_{gb}(x_i) = -3.252$$

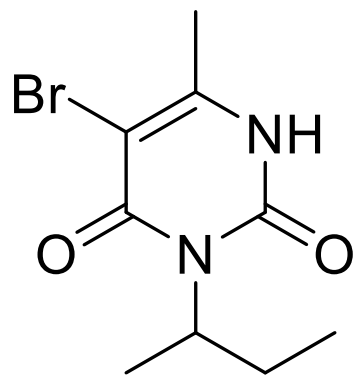
Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 3: Compute the baseline prediction

$$X = x_{\text{train}}$$

$$\varphi_0 = \mathbb{E}[f(X)] = -2.885$$



Bromacil

$$f_{gb}(x_i) = -3.252$$

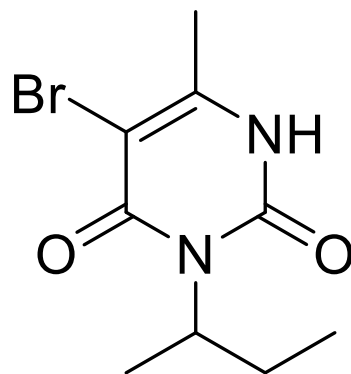
Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 4: Consider all feature coalitions without molecular weight

$$X = \mathbf{x}_{\text{train}}$$

$$\varphi_0 = \mathbb{E}[f(X)] = -2.885$$



Bromacil

$$f_{gb}(x_i) = -3.252$$

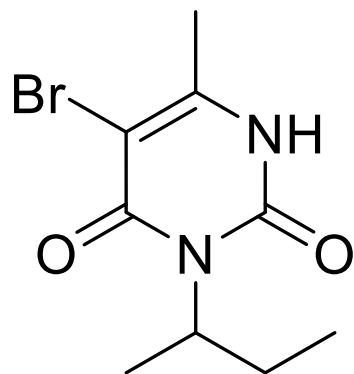
{
{Polar Surface Area}
{Number of Rings}
{Polar Surface Area, Number of Rings}
{Number of Rings, Number of Rotatable Bonds}
...
{Polar Surface Area, Number of Rings, Rotatable Bonds, H-Bond Donors, Minimum Degree}

$$2^{6-1} = 32 \text{ total coalitions}$$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 5: Measure what happens when molecular weight is added



Bromacil

$$f_{gb}(x_i) = -3.252$$

$$X = X_{\text{train}}$$

$$\varphi_0 = \mathbb{E}[f(X)] = -2.885$$

{
{Polar Surface Area}
{Number of Rings}
{Polar Surface Area, Number of Rings}
{Number of Rings, Number of Rotatable Bonds}
...
{Polar Surface Area, Number of Rings, Rotatable Bonds, H-Bond Donors, Minimum Degree}

$$2^{6-1} = 32 \text{ total coalitions}$$

Compute $v(S \cup \{\text{MW}\}) - v(S)$ for each subset S

E.g.,

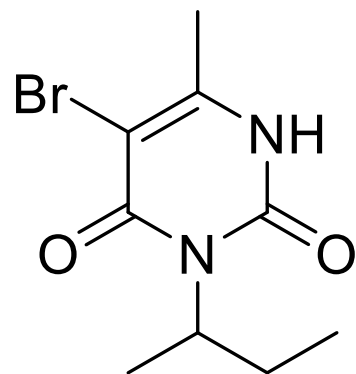
$$v(\{\text{Rings}, \text{MW}\}) - v(\{\text{Rings}\}) = -1.02$$

$$v(\{\text{PSA}, \text{Rings}, \text{MW}\}) - v(\{\text{PSA}, \text{Rings}\}) = -0.95$$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 6: Average all marginal contributions with Shapley weights



Bromacil

$$f_{gb}(x_i) = -3.252$$

$$X = X_{\text{train}}$$

$$\varphi_0 = \mathbb{E}[f(X)] = -2.885$$

{
 {Polar Surface Area}
 {Number of Rings}
 {Polar Surface Area, Number of Rings}
 {Number of Rings, Number of Rotatable Bonds}
 ...
 {Polar Surface Area, Number of Rings, Rotatable Bonds, H-Bond Donors, Minimum Degree}

$$2^{6-1} = 32 \text{ total coalitions}$$

Compute $v(S \cup \{\text{MW}\}) - v(S)$ for each subset S

E.g.,

$$v(\{\text{Rings}, \text{MW}\}) - v(\{\text{Rings}\}) = -1.02$$

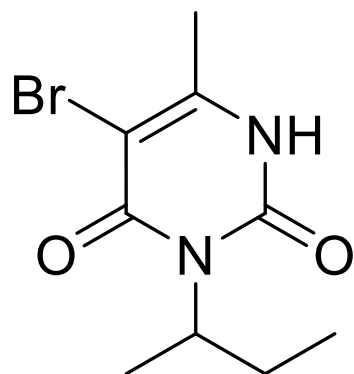
$$v(\{\text{PSA}, \text{Rings}, \text{MW}\}) - v(\{\text{PSA}, \text{Rings}\}) = -0.95$$

$$\begin{aligned} \varphi_{\text{MW}} &= \sum_{S \subseteq M \setminus \{\text{MW}\}} \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{\text{MW}\}) - v(S)] \\ &= -1.01 \end{aligned}$$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

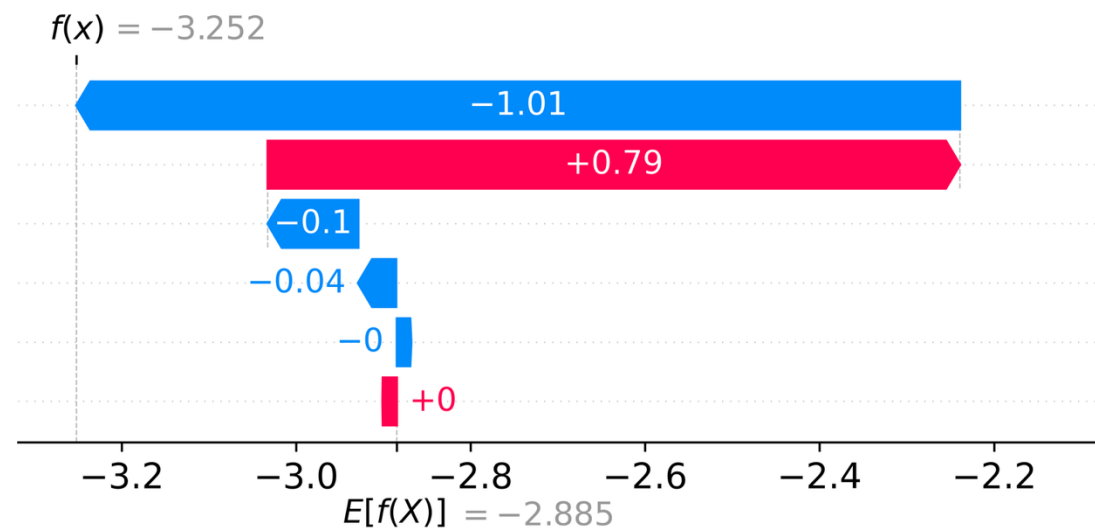
Example:

$$f_{gb}(x_i) = -3.252$$



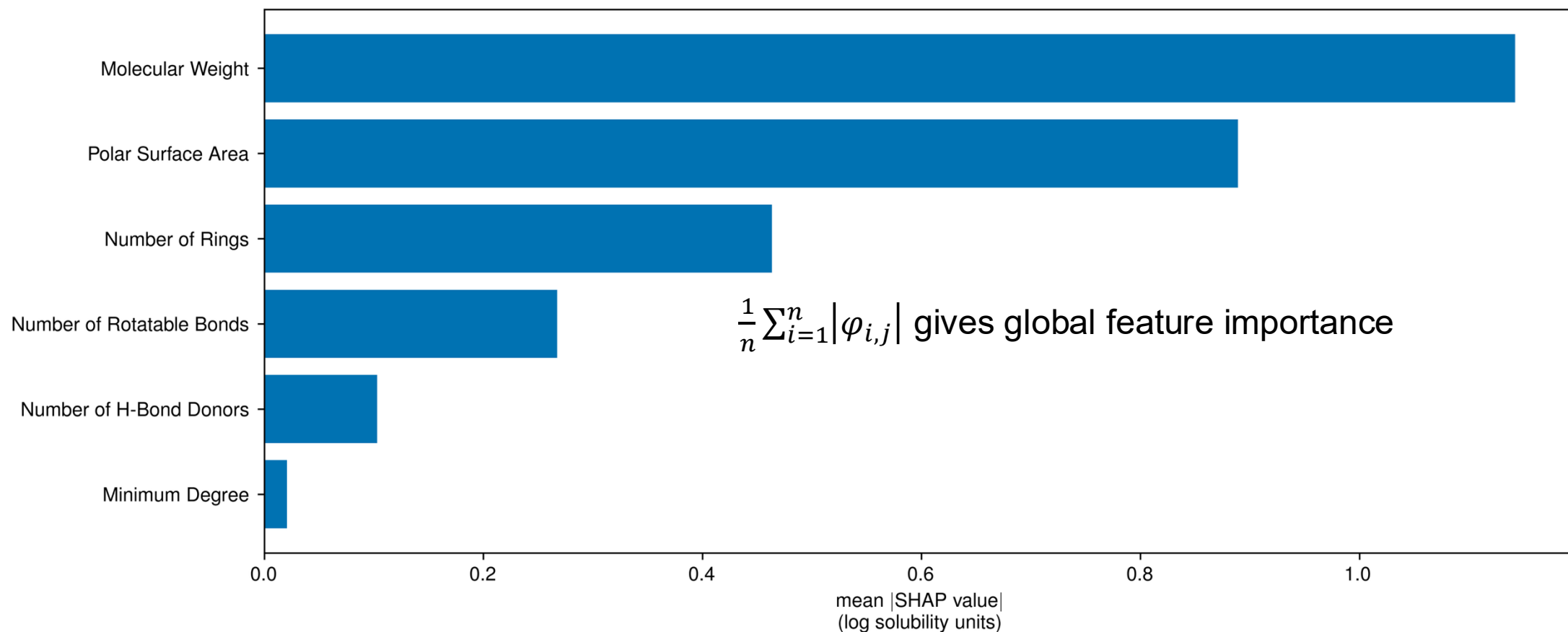
Bromacil

261.119 = Molecular Weight
54.86 = Polar Surface Area
1 = Number of Rings
1 = Number of H-Bond Donors
1 = Minimum Degree
2 = Number of Rotatable Bonds

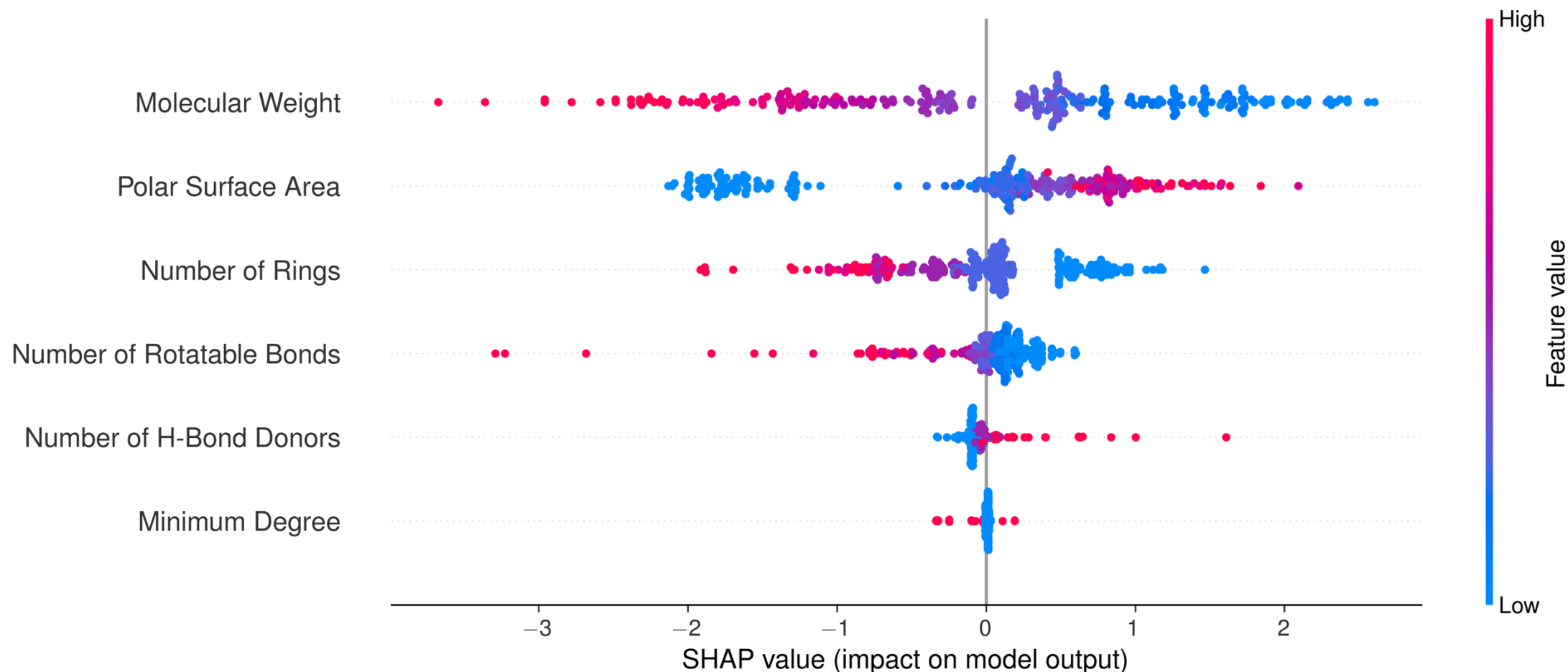


How much did molecular weight, polar surface area, ring count, and the other descriptors push the predicted log solubility above or below a reference prediction?

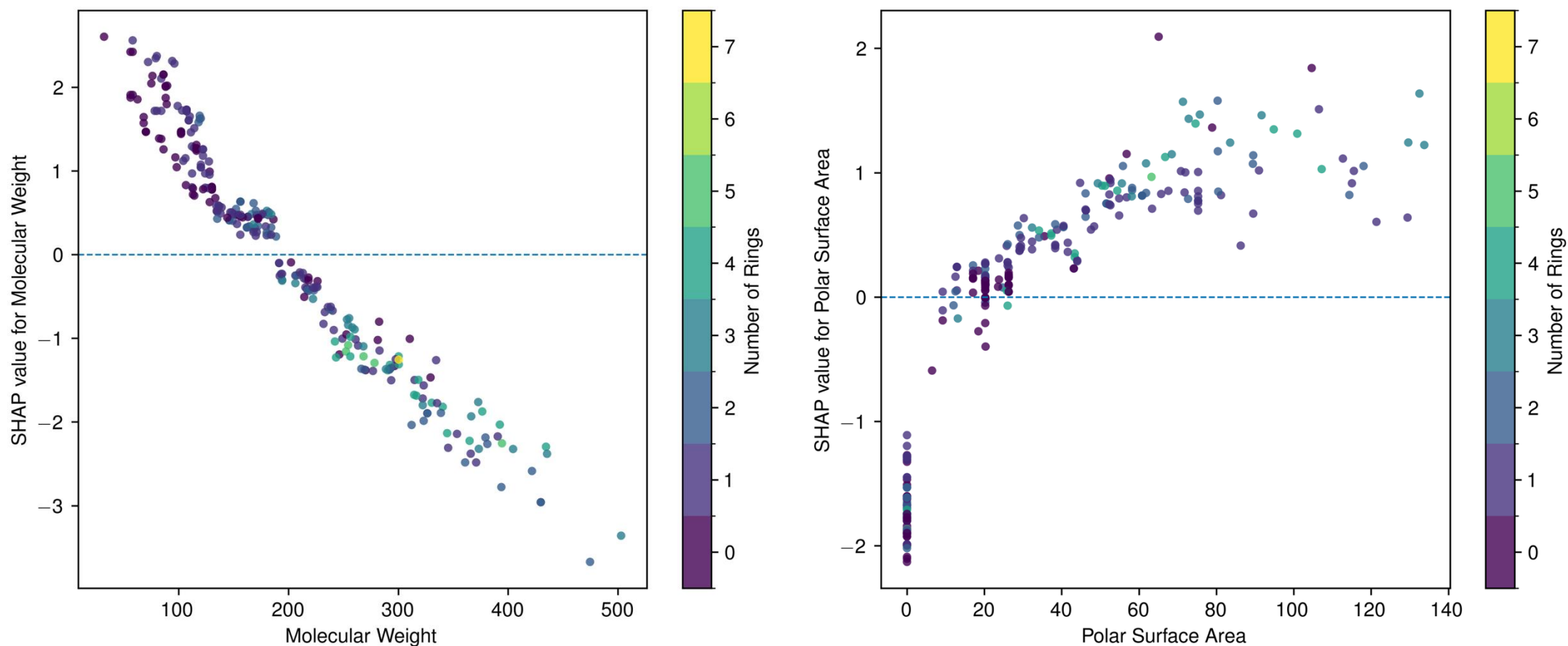
Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)



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Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Strengths

- **Provides local additive feature attributions** for individual predictions
- **Works well when** you want both local explanations and global summaries from aggregated SHAP values
- **Helps reveal** direction, magnitude, and distribution of feature contributions across samples

Limitations

- **Does not prove mechanism or causality**; it explains the model's attribution of a prediction
- **Sensitive to correlated feature and background/reference distribution**

Closing thoughts

Choosing the right model

Question	Method
Which features matter globally?	MDI, permutation importance, mean SHAP
How does a feature affect predictions?	ICE/PDP
Why this specific prediction?	LIME, SHAP
Does the model learn chemistry or shortcuts?	Compare methods + domain knowledge

A practical workflow

1. Start with an interpretable baseline
2. Compare against flexible models
3. Use global explanation methods to identify model reliance
4. Use local explanations to inspect examples and outliers
5. Check whether explanations make chemical sense

Common failure modes

- correlated descriptors split or hide importance
- local explanations are treated as global rules
- feature attributions are interpreted as mechanisms
- visually convincing plots create false confidence

Additional resources

Accumulated local effects (ALE)

Feature-effect plots that reduce extrapolation issues when features are correlated.

[Apley & Zhu, JRSS B 2020](#)

Explainable Boosting Machines

Interpretable GAM-like models with learned nonlinear feature functions and interactions.

interpret.ml/docs/ebm.html

GNNExplainer

Subgraph and node-feature explanations for graph neural networks.

arxiv.org/abs/1903.03894

Use interpretable models when possible

A cautionary perspective on post-hoc explanations in high-stakes settings.

[Rudin, Nat. Mach. Intell. 2019](#)

Counterfactual explanations

Ask: “What minimal molecular/property change would flip the prediction?”

interpret.ml/DiCE

Integrated gradients / Captum

Attribution methods for neural networks; useful for differentiable molecular models.

captum.ai

Sanity checks for explanations

Test whether saliency/explanation maps actually depend on the model and data.

[Adebayo et al., NeurIPS 2018](#)

Benchmark explanation quality

Libraries such as Quantus help compare robustness and faithfulness of explanations.

quantus.readthedocs.io